SU(2) on the full mid-plane: the relative-order-t halving defect

Result, 2026-09-27 (formal one-loop calculation). For one halving of direction 1, with isotropic coarse heat time $t = \lambda_3 a$, the zero-momentum coupling coefficients in Hypothesis P(α) are

$$\boxed{\begin{aligned}\delta_t^{12} = \delta_t^{13} &= \frac{t}{24}\int_{\mathcal{B}} \frac{q}{R} + O(t^2) > 0,\\ \delta_t^{23} &= t\left(-\frac{1}{12} + 2\mathcal{I}\right) + O(t^2) < 0,\end{aligned}} \tag{1}$$

where the inequalities describe the leading coefficients, and

$$\int_{\mathcal{B}} := \int_{-\pi}^{\pi}\int_{-\pi}^{\pi} \frac{dk_2\, dk_3}{(2\pi)^2}, \qquad q = 2 - 2\cos k_2, \quad r = 2 - 2\cos k_3, \quad d = 8 + r, \quad R = 8 + r + q,$$

$$\boxed{\mathcal{I} = \int_{\mathcal{B}} \left\{ \frac{r^3 + 10r^2 + 4r - 1248 + q(8+r)^2}{24(8+r)^2(8+r+q)} - \frac{q(2\sin k_3)^2(2\sin k_2)^2}{4(8+r)(8+r+q)^4} \right\} < 0.} \tag{2}$$

All integrals are finite. Their denominators are bounded below by positive powers of 8. The first numerator in (2) increases with each of $q, r \in [0, 4]$ and has maximum -432; the second summand is nonpositive. The cut integral is strictly positive. These statements use no numerical evaluation.

The cut shifts come from the normalized bridge-curvature determinant (N2). For equal adjacent layers, the bridges have an exactly Gaussian zero-image density in exponential coordinates. The transverse shift then consists of the mid-face heat-kernel amplitude, $-t/12$, and the coupled noncommutative fluctuation determinant, $2t\mathcal{I}$ (N3, including the curvature of the mid-face action). Neighbouring faces share fluctuations throughout this calculation.

After these shifts and the vacuum term are extracted, the formal **bulk local** smooth-field expansion contains only operators of geometric dimension at least six. Here $[D] = 1$ and $[F] = 2$, the convention used for irrelevance in the series/parallel note. The full small-field statement still requires the volume-uniform, normalized estimates in §7. Equation (1) gives bulk coefficients; finite tori also carry the explicit flat-holonomy term (16) below. It obstructs the literal flux-only P(α) inequality at fixed torus size as $t \to 0$, and is exponentially small when the number of sites around the torus tends to infinity. Iteration requires a further stability estimate.

Inputs and scope. The exact definition of Ψ and $\mathcal{D}$ is Proposition 1, equations (3)–(4), and Hypothesis P(α) of the series/parallel note (full-read: relevant sections, including Propositions 4, 6 and 7). The bridge checks use the exact midpoint note, Theorems 1–4 (full-read). The calculations below are written derivations of the formal coefficients; uniformity in the spin sum is a separate estimate (§7). We use $[T_a, T_b] = \epsilon_{abc}T_c$, $|T_a| = 1$, $C_2(\mathrm{adj}) = 2$, and normalized Haar measure.

1. Expansion and the massive mid-plane covariance

Use the mid-vertex gauge of Proposition 1 and write $m_e = m_{*e}\exp\xi_e$. At the trivial background the quadratic exponent is

$$\frac{1}{t}\left(2\sum_e |\xi_e|^2 + \frac{1}{4}\sum_g |\bar{\Phi}_g + (C\xi)_g|^2\right), \qquad \bar{\Phi} = \frac{\Phi_\ell + \Phi_{\ell+1}}{2}.$$

The Fourier symbol of the curl and the inverse fluctuation Hessian are

$$c(k) = (1 - e^{ik_3},\ e^{ik_2} - 1), \qquad H_0 = 4I + \frac{1}{2}c^*c, \qquad G = H_0^{-1} = \frac{1}{4}\left(I - \frac{c^*c}{R}\right). \tag{3}$$

There is an additional identity matrix on the three colour components. In particular $4I \le H_0 \le 8I$, including zero momentum. The bridge term supplies a mass for every mid-edge fluctuation. Integrating (3) gives precisely the free defect $\mathcal{D}_0$ of Proposition 2; the plane coupling agrees with the covariance in Proposition 4.

For precision about orders, introduce local boundary coordinates X and the zero-image exponent $\mathcal{S}(X,\xi)$ of the *normalized* integral defining the defect. Thus it includes the subtraction $-\sum_e |X_e|^2/2$ from the bridge denominators and $-\sum_{f\perp 1} |X_f|^2/4$ from the old-face ratio. Its positive terms are the two squared half-face distances per cut face and $\sum_g |\log m_{\partial g}|^2/4$. Let $F(X) = \min_\xi \mathcal{S}(X,\xi)$. The formal normalized Laplace expansion of the bulk part is

$$\mathcal{D}(X,t) = c_{\mathrm{vac}}(t)N + \frac{F_2+F_3+F_4}{t} + A_2(X) + \text{higher terms}, \qquad F_2/t = \mathcal{D}_0. \tag{4}$$

Finite periodic planes also have the winding functional of §7. Subscripts are homogeneous boundary degree about the trivial field. The one-loop amplitude A includes $\frac{1}{2}\log\det H$, Haar and heat-kernel amplitudes, and the bridge denominators. An explicit prescription, useful also for its derivative terms, is

$$\begin{aligned} A(X) = {} & \tfrac{1}{2}\log\det H(X) + \tfrac{1}{2}\sum_e [\log j(L_e^-) + \log j(L_e^+) - \log j(X_e)] \\ & - \sum_e \log j(\xi_e^*) + \tfrac{1}{2}\sum_g \log j(Y_g^*), \qquad j(Z) = \left[\frac{\sin(|Z|/2)}{|Z|/2}\right]^2. \end{aligned}$$

Here ξ^* is the classical saddle, $L_e^\pm$ are the logarithms of its two half-face holonomies, Y_g^* is its mid-face logarithm, and H is the Hessian in the ξ coordinates. Boundary-independent constants are suppressed. Gauge invariance removes the linear term of the bulk local expansion. Its quadratic term is

$$A_2(X) = \frac{1}{48}\int_{\mathcal{B}} \frac{q}{R} \sum_{f\in(12),(13)} |X_f|^2 + \left(-\frac{1}{24} + \mathcal{I}\right) \sum_{f\in(23)} |X_f|^2 + A_{2,\mathrm{der}}(X). \tag{5}$$

The last quadratic kernel has zero constant term in its covariant external-momentum expansion. Equation (5) defines the convention for coupling shifts: the coefficient of $|X_f|^2$ is $\delta_t^{\mu\nu}/(2t)$. On a finite periodic plane, momentum integrals for local contractions become sums; (1)–(2) specify their bulk limit.

For typical perturbative counting $X = O(\sqrt{t})$, F_3/t has order $\sqrt{t}$, and both F_4/t and A_2 have order t. Keeping the cubic and quartic classical terms is essential to an order-t expansion. On P(α)'s larger set, $|X| \le t^{1/2-\delta}$, retain their actual powers of X rather than replacing X^2 by t.

2. N2: curvature of the normalized bridges

Proposition 6 in its group-general form gives

$$h_X = I + \frac{(\mathrm{ad}\, X)^2}{48} + O(|X|^4), \qquad \mathrm{Cov}\,\xi = \frac{t}{4}h_X^{-1} + O_X(t^2). \tag{6}$$

For $SU(2)$, h_X has eigenvalues $1, h, h$, with $h = (|X|/4)\cot(|X|/4)$. The trace of its covariance excess is $t|X|^2/96$, or $t|X|^2/288$ per colour direction. This is the softening in Proposition 6, with its axial part retained.

Theorems 1–2 of the midpoint note check (6) in every fixed-spin character. Theorem 3 gives the scalar spin-1/2 matrix; Theorem 4 gives distinct spin-1 entries and their image bounds. At $X = 0$ there is a stronger useful identity. In exponential coordinates the zero-image factors satisfy

$$k_{t/2}(e^{\xi})^2\,dm \;\propto\; e^{-2|\xi|^2/t}\,d^3\xi. \tag{7}$$

Indeed, the two factors $(|\xi|/2)/\sin(|\xi|/2)$ cancel the Haar Jacobian $[\sin(|\xi|/2)/(|\xi|/2)]^2$. Continuing this Gaussian to $\mathbb{R}^3$ gives covariance exactly $tI/4$; truncation to the injectivity chart changes it by an exponentially small tail. Its fundamental expectation $e^{-t/32}(1-t/16)$ and spin-1 expectation $[1+2e^{-t/8}(1-t/4)]/3$ agree with the exact endpoint limits. In these coordinates the central-bridge covariance is exactly $tI/4$ up to the exponentially small tail. (Convention: §1 writes $m_e = m_* e^{\xi}$ and §3 transports as $m_e = e^{\xi}U_e$; the two lists differ by a diagonal unitary, which leaves every determinant unchanged. Abelian limit: $(\operatorname{ad} X)^2 = 0$, no commutator and no Jacobian factor, so all three shifts vanish, matching the free-field check of P(α) and Propositions 2 and 4 of the series/parallel note. Fable referee, 2026-09-27.)

To extract the 12 coefficient, take constant commuting cut flux $X_{12} = X$, $X_{13} = 0$, and flat midpoint connection. Adjacent layers can be represented by constant links $e^{-X/2}$ and $e^{X/2}$ in direction 2. The classical defect vanishes. The Hessian in (3) changes by $4(h_X - I)$ on direction-2 edges, while bridge normalization supplies $-\frac{1}{2}\operatorname{Tr}\log h_X$. Therefore its quadratic amplitude is

$$\begin{aligned} A_{2,\mathrm{cut}} &= \frac{1}{96}\operatorname{Tr}[(4G - I)(\operatorname{ad} X)^2] \\ &= \frac{|X|^2}{48}\int_{\mathcal{B}} \frac{r}{R} = \frac{|X|^2}{48}\int_{\mathcal{B}} \frac{q}{R}. \end{aligned} \tag{8}$$

Here $4G_{22} - 1 = -r/R$ and $\operatorname{tr}_{\mathrm{colour}}(\operatorname{ad} X)^2 = -2|X|^2$. The exchange of the two integration variables gives the last equality. Direction 3 gives the same result. Normalization explains the positive sign: the unconstrained bridge's curvature determinant is subtracted, and the mid-face weights constrain its broadened directions.

N2 also contributes higher boundary-degree terms. For example, at the Gaussian saddle $\xi^{(0)}$ the quadratic bridge-curvature insertion in the classical exponent is

$$\frac{1}{24t}\sum_e \langle \xi_e^{(0)}, (\operatorname{ad} X_e)^2 \xi_e^{(0)}\rangle = -\frac{1}{24t}\sum_e |[X_e, \xi_e^{(0)}]|^2. \tag{9}$$

It belongs to F_4/t; its coefficient follows directly from (6). Other fourth-degree classical terms are retained by the saddle formula in §6. At equal adjacent layers $X_{12} = X_{13} = 0$, (7) removes the bridge-shape contribution altogether, which isolates the transverse calculation below.

3. N3: the full plaquette Hessian

Set both adjacent layers equal to a constant Abelian magnetic background, with $U_{23} = e^{BT_3}$ and zero cut flux. It is sufficient to extract the local bulk coefficient at $B = 0$; one can use a large box before taking this coefficient. The midpoint connection is an exact stationary point, since neighbouring plaquettes have the same covariantly constant flux. Its bridge coordinates have density (7).

For four oriented fluctuations transported to a plaquette base, write $\eta_1, \dots, \eta_4$, and $Y = BT_3$. The quadratic part of $L = \frac{1}{2}|\log m_{\partial g}|^2$ is

$$L^{(2)} = \frac{1}{2}\left\langle \sum_i \eta_i, Q_Y \sum_i \eta_i \right\rangle + \frac{1}{2}\sum_{i<j}\langle Y, [\eta_i, \eta_j]\rangle, \qquad Q_Y = \frac{\operatorname{ad} Y}{2}\coth\frac{\operatorname{ad} Y}{2}. \tag{10}$$

This follows by inserting $\log(\prod_i e^{\eta_i}) = \sum_i \eta_i + \frac{1}{2}\sum_{i<j}[\eta_i, \eta_j] + O(\eta^3)$ into the squared-distance function. On the charged colour plane $Q_Y = h_B I$, $h_B = (B/2)\cot(B/2)$. The neutral colour has the unchanged free Hessian. Both the radial Hessian and the ordered commutator in (10) are needed.

Here is an explicit determinant evaluation retaining both terms. Choose background links $U_2 = 1$, $U_3(x) = e^{Bx_2T_3}$ and Fourier transform in direction 3. Combine the two charged real colours into one complex field (u, v) on edges in directions $(2, 3)$. Put

$$s_B = \frac{B/2}{\sin(B/2)}, \qquad h_B = s_B \cos(B/2), \qquad z = k_3 + Bx_2, \quad y = z + B/2,$$

and let E shift x_2 by one. The charged Hessian of L is the Hermitian block operator K with

$$\begin{aligned} K_{uu} &= 2h_B - 2s_B \cos y, \\ K_{vv} &= 2h_B - s_B(e^{-iB/2}E + e^{iB/2}E^*), \\ K_{uv} &= f(y)E + g(y), \\ f(y) &= s_B(e^{-iB/2} - e^{-iy}), \qquad g(y) = s_B(e^{-iy} - e^{iB/2}). \end{aligned} \tag{11}$$

For example the four transported charged fluctuations are $u(x)$, $v(x+\hat{2})$, $-e^{iB(x_2+1)}u(x+\hat{3})$, $-e^{iB}v(x)$. Substitution in (10), using $h_B \pm iB/2 = s_B e^{\pm iB/2}$, gives (11).

The fluctuation Hessian in the exponent is $H = 4I + K/2$. The two charged real colours contribute $\frac{1}{2}\operatorname{Tr}_{\mathbb{R}} \log H = \operatorname{Tr}_{\mathbb{C}} \log H$. Thus the desired one-loop coefficient is the B^2 coefficient per site of $\log\det(8 + K)$. The neutral determinant is constant. This formula contains the connected two-vertex contribution $-\frac{1}{4}\operatorname{Tr}(H_0^{-1}H_1H_0^{-1}H_1)$ in real-colour notation, together with $\frac{1}{2}\operatorname{Tr}(H_0^{-1}H_2)$; it keeps their cancellations in one expression.

4. Evaluating the transverse coefficient as a lattice integral

The uu block of $8 + K$ is multiplication by

$$D(y) = 8 + 2s_B[\cos(B/2) - \cos y].$$

Eliminating u gives a scalar tridiagonal Schur complement S. The elementary identity $g(y)^* f(y) = -s_B e^{-iB/2}[D(y) - 8]$ gives its forward hopping $-8s_B e^{-iB/2}/D(y)$. Its diagonal at site x_2 is

$$8 + 2h_B - s_B^2 \left[\frac{2 - 2\cos(z + B)}{D(z + B/2)} + \frac{2 - 2\cos(z - B)}{D(z - B/2)}\right]. \tag{12}$$

A constant unitary phase removes $e^{-iB/2}$ from the hopping. Consequently $\log\det(8 + K) = \operatorname{Tr}\log D + \operatorname{Tr}\log S$ exactly at finite volume with compatible boundary conditions.

For the bulk derivative expansion let $k = k_2$, $r = 2 - 2\cos z$, $q = 2 - 2\cos k$, $d = 8 + r$, $R = d + q$, and $w = \sin^2 z = r(4 - r)/4$. The scalar symmetric, or Weyl, symbol of S has expansion $s_0 + B^2 s_2 + O(B^4)$, where

$$\begin{aligned} s_0 &= \frac{8R}{d}, \\ s_2 &= -\frac{1}{6} - \frac{2\cos z}{d} + \frac{4w}{d^2} - \frac{2rw}{d^3} - \frac{r^2}{6d^2} - \frac{r}{6d} - \frac{28\cos k}{3d^2}. \end{aligned} \tag{13}$$

Indeed $D(z \pm B/2) = d \pm B\sin z - B^2 r/12 + O(B^3)$, while at a hopping midpoint $D(y) = d(y) + B^2(r(y)/24 - 1/4) + O(B^4)$. These two expansions in (12) yield (13). The multiplication determinant contributes $\int_{\mathcal{B}}(r - 6)/(24d)$ at order B^2.

The trace of the logarithm also includes the symbol-product correction. Since z and k have commutator of magnitude B, the second-order product of scalar symbols is

$$f \star g = fg + \frac{iB}{2}\{f, g\} - \frac{B^2}{8}(f_{zz}g_{kk} - 2f_{zk}g_{zk} + f_{kk}g_{zz}) + O(B^3).$$

Expanding the resolvent of s_0+u in this formula and integrating $\log s_0 = \int_0^\infty [(1+u)^{-1} - (s_0+u)^{-1}]\,du$ gives

$$[B^2]\,\mathrm{Tr}\log S\,/N = \int_{\mathcal{B}} \left[\frac{s_2}{s_0} + \frac{\det(\partial^2 s_0)}{24 s_0^2}\right]. \tag{14}$$

Here and below $[B^2]$ means the Taylor coefficient. To check the factor $1/24$, the resolvent correction is $-\det(\partial^2 s_0)/(4(s_0+u)^3) + Q/(4(s_0+u)^4)$, with $Q = s_{0,zz}s_{0,k}^2 - 2s_{0,zk}s_{0,z}s_{0,k} + s_{0,kk}s_{0,z}^2$. Integration in u gives $\det(\partial^2 s_0)/(8s_0^2) - Q/(12 s_0^3)$. Integration by parts on the momentum torus gives $\int Q/s_0^3 = \int \det(\partial^2 s_0)/s_0^2$, proving (14).

Two final simplifications expose the sign. Direct substitution gives

$$\frac{r-6}{24d} + \frac{s_2}{s_0} = \frac{r^3+10r^2+4r-1248+qd^2}{24d^2R}.$$

For $s_0 = 8(1+q/d)$, integration by parts first in z and then in k gives

$$\int_{\mathcal{B}} \frac{\det(\partial^2 s_0)}{s_0^2} = -6\int_{\mathcal{B}} \frac{q(r')^2(q')^2}{dR^4}.$$

Together these are exactly (2). As an algebraic normalization check, replacing the 8 in $8+K$ by a large mass parameter M gives $[B^2]\,\mathrm{Tr}\log(M+K)/N = -1/(3M) + 1/(6M^2) + O(M^{-3})$: (11) yields $[B^2]\,\mathrm{Tr}\,K/N = -1/3$ and $[B^2]\,\mathrm{Tr}\,K^2/N = -1/3$ directly.

Finally, the zero-image heat kernel contains $j(Y)^{-1/2} = (|Y|/2)/\sin(|Y|/2)$, so the mid-face amplitude contributes $-\log j(Y)^{-1/2} = -|Y|^2/24 + O(|Y|^4)$ to the action. The corresponding factors in $k_{2t}(U_f)/k_t(U_f)$ cancel exactly. Adding this $-1/24$ to $\mathcal{I}$ and multiplying by $2t$ proves (1)'s transverse coefficient in the formal expansion.

5. What the midpoint matrices say about N3

The exact midpoint results provide a local check and specify their scope. In spin 1/2, the centered expectation is a scalar matrix, so an isolated cube's mid-face factors into four scalar bridge expectations times its interpolated holonomy. In spin 1, Theorem 4 gives

$$\lambda_0 - \lambda_1 = \frac{t}{8}(1-1/h) + O_X(t^2) = -\frac{t|X|^2}{384} + O(t|X|^4) + O_X(t^2).$$

The unequal eigenvalues retain the axis of each cut flux. Their commutator is proportional to $[P_i, P_j]$ for transported axial projectors, and therefore survives for generic oblique axes. This fixed-spin matrix commutator has order t^2 at fixed side angles, so the full order-t plane coefficient needs the plane Hessian of §3.

The heat-kernel character sum samples spins of size $t^{-1/2}$, and each shared edge joins neighbouring face representations. The local-coordinate Hessian (10)–(14) resums the needed quadratic fluctuations before any fixed-spin truncation. Its linear-in-Y commutator vertices have a connected square at quadratic boundary degree. This is the N3 contribution relevant to a coupling shift. Theorems 1–4 test the curvature term N2 and the bridge law at $X=0$ ($\lambda(0,t) = 1 - t/4$), and the scalar factorization of the isolated fundamental sector; they give no check on the value of $\mathcal{I}$.

For a more detailed split of the determinant, the radial-curvature part $Q_Y - I = (\mathrm{ad}\,Y)^2/12 + O(Y^4)$ contributes

$$\mathcal{I}_{\mathrm{radial}} = -\frac{1}{12}\int_{\mathcal{B}} \frac{q+r}{R}.$$

The rest is $\mathcal{I} - \mathcal{I}_{\mathrm{radial}}$, including parallel transport and ordered-commutator vertices and their connected square. Equations (10)–(14) specify this split unambiguously. The strict negative

sign in (2) concerns the complete determinant. The separate bridge term N2 is (8); changing fluctuation coordinates can redistribute intermediate geometric and product terms while leaving (1) fixed.

6. The other terms and their smooth-field dimensions

The classical part in (4) can be retained without guessing individual local counterterms. Expand the explicit squared-distance exponent of §1 as $\mathcal{S}_2 + \mathcal{S}_3 + \mathcal{S}_4 + \cdots$ and solve its Gaussian saddle, $\xi^{(0)} = -GC^*\bar{\Phi}/2$. Then

$$\begin{aligned} F_3 &= \mathcal{S}_3(X, \xi^{(0)}), \\ F_4 &= \mathcal{S}_4(X, \xi^{(0)}) - \frac{1}{2}\langle \partial_\xi \mathcal{S}_3(X, \xi^{(0)}), G\, \partial_\xi \mathcal{S}_3(X, \xi^{(0)})\rangle. \end{aligned} \tag{15}$$

Thus (4), (5), (10) and (15) define every term through the stated boundary and loop orders. This includes (9), the classical interpolation commutator, and the quartic terms from the ordered mid-face product. A smooth-field example is the N1 term $\langle F_{23}, [F_{12}, F_{13}]\rangle$, of dimension six; quartic curvature invariants have dimension eight. The free defect and the quadratic remainder $A_{2,\mathrm{der}}$ start with covariant derivatives of curvature, of dimension six.

The reason this list exhausts relevant smooth-field terms is local. The inverse of (3) has exponential decay in lattice distance, so its formal connected contractions have analytic external-momentum expansions. Gauge invariance forces their bulk local terms to be curvature invariants. Reflection symmetry forbids the odd-parity Chern–Simons term and dimension-five terms: a dimension-five term has five index slots, so some coordinate direction occurs an odd number of times and the reflection of that direction reverses its sign. At dimension four, independent reflections leave only $|F_{12}|^2$, $|F_{13}|^2$ and $|F_{23}|^2$; their coefficients are (1). Differences between opposite faces and mixed quadratic kernels, allowed by Proposition 7, give higher derivatives in this expansion. Terms with three curvatures start at dimension six; for $SU(2)$ the only cubic is the N1 term, since $\langle A, [A, B]\rangle = 0$, while for $SU(3)$ the symmetric tensor d_{abc} admits a second dimension-six cubic, $\mathrm{tr}(F_{12}\{F_{13}, F_{23}\})$. This argument uses smooth fields and the bulk local expansion; arbitrary small plaquettes can still vary at lattice momenta of order one.

Consequently the mid-plane extension of Proposition 7 has the expected **formal** operator content. Equations (1)–(15) leave no extra relevant bulk operator at relative order t. They leave a quantitative uniform-integral problem, specified next.

7. A finite-torus obstruction and the estimates needed for a theorem

The global qualification has a direct test. On an $N_2 \times N_3$ periodic plane take identical flat adjacent layers with commuting holonomies $e^{\theta_2 T_3}$ and $e^{\theta_3 T_3}$. Every coarse plaquette logarithm vanishes. The charged version of (3) has momenta $k_j + \theta_j/N_j$, where $k_j = 2\pi n_j/N_j$. Finite-dimensional Laplace expansion at its unique zero-action saddle therefore gives

$$\begin{aligned} \mathcal{D}(U_\theta, t) - \mathcal{D}(1, t) &= \Omega_N(\theta) + O_N(t), \\ \Omega_N(\theta) &= \sum_k \log \frac{12 - 2\cos(k_2 + \theta_2/N_2) - 2\cos(k_3 + \theta_3/N_3)}{12 - 2\cos k_2 - 2\cos k_3}. \end{aligned} \tag{16}$$

The bridge distance term has a unique minimum and positive Hessian for each such fixed flat background. Rescaling the finitely many normal coordinates by $\sqrt{t}$ gives (16); the complement of a fixed saddle neighbourhood has exponentially small weight. The argument is a fixed-N Laplace expansion.

There is a signed, elementary bound. Write $N = N_2 N_3$ and $n_* = \min(N_2, N_3)$. Expand $\log(12 - \mathsf{A}_\theta)$, with A_θ the nearest-neighbour adjacency matrix carrying the flat phases. If $a_n(w)$ counts length-n walks from one site to its translate by $(w_2 N_2, w_3 N_3)$ on the covering lattice, then

$$\begin{aligned}\Omega_N(\theta) &= N\sum_{n\geq 1}\frac{1}{n12^n}\sum_{w\neq 0}a_n(w)[1-\cos(w\cdot\theta)],\\ 0\leq\Omega_N(\theta) &\leq \frac{3N}{n_*}\,3^{-n_*}.\end{aligned}\tag{17}$$

The bound uses $\sum_w a_n(w)\leq 4^n$ and $n\geq n_*$ for nonzero winding. Straight winding paths make the inequality strict whenever at least one adjoint holonomy is nontrivial. At fixed N, (16) is an order-one background-dependent term as $t\to 0$. Both coupling shifts and every local curvature operator vanish on these configurations. Subtracting $\mathrm{P}(\alpha)$ for U_θ and for 1 would bound their difference by $Ct^\alpha N$, contradicting (16). So $\mathrm{P}(\alpha)$ as literally stated fails on fixed periodic planes with arbitrary flat holonomies. The failure is an artifact of $t\to 0$ at fixed N: under the refinement scaling, with $\min_j N_j\geq c_0/t$ (fixed physical size), the winding term is $O(t^{-1}e^{-c/t})$ and lies inside every $\mathrm{P}(\alpha)$ remainder. The simplest repair is to add the clause $\min_j N_j\geq c_0/t$ to $\mathrm{P}(\alpha)$. The exponential smallness comes from the massive mid-plane modes only; the coarse theory's own holonomy potential is a separate object, not exponentially small (Fable referee, 2026-09-27).

For $N_j = L_j/a$ at fixed positive physical lengths and $t=\lambda_3 a$, (17) is beyond every power of t. Its origin is a winding determinant; assigning it a local operator dimension would lose its dependence on the torus. This is the precise exception to the bulk irrelevance statement of §6.

A sufficient formulation for a theorem is the following. Work in local gauge charts on the small-field set, including the treatment of global holonomies and boundary conditions. For every plane size require a unique contributing small saddle, a Hessian bounded below by a fixed $cI>0$, and exponentially summable connected kernels for its classical value F and normalized one-loop amplitude A. Constants and decay rates must be independent of plane area. The normalized integral, including all bridge denominators, Haar factors and images, must obey the following after separating a finite-volume winding functional $W_N(U,t)$ (absent in the infinite-plane bulk expansion):

$$\begin{aligned}\mathcal{D}(U,t) &= c_{\mathrm{vac}}(t)N + F(U)/t + A(U) - A(1) + W_N(U,t) + E(U,t),\\ |E(U,t)-E(1,t)| &\leq Ct\sum_p |X_p|^2 + CNe^{-c/t},\\ |F-F_2-F_3-F_4| &\leq C\varepsilon^3\sum_p |X_p|^2,\\ |A-A(1)-A_2| &\leq C\varepsilon\sum_p |X_p|^2, \qquad \varepsilon=\sup_p|X_p|\leq t^{1/2-\delta}.\end{aligned}\tag{18}$$

An $E(1,t)$ constant is absorbed into c_{vac}. The quadratic Taylor kernel of A must have the bulk coefficients (5), with exponentially summable second spatial moments. Here F, A denote the bulk local kernels; W_N collects their winding corrections and has $W_N(U_\theta,t)=\Omega_N(\theta)+O_N(t)$ after vacuum subtraction. One further needs $|W_N|\leq Ce^{-\kappa n_*}\sum_p(1+|X_p|^2/t)$ with fixed $\kappa>0$ in the chosen small-field charts. These conditions justify both the order-t coefficient extraction and the derivative expansion. They also give, after retaining the terms in (4), a remainder bounded by

$$C\left(\frac{\varepsilon^3}{t}+\varepsilon+t\right)\sum_p|X_p|^2 + CNe^{-c/t}.$$

For the weaker unperturbed $\mathrm{P}(\alpha)$ inequality, it suffices to replace the strong remainder line of (18) by $|E|\leq Ct\sum_p(1+|X_p|^2/t)$ and use $|F-F_2|\leq C\varepsilon\sum_p|X_p|^2$ and $|A-A(1)|\leq C\sum_p|X_p|^2$. Then $\alpha=1/2-\delta>2\delta$ for $0<\delta<1/6$, exactly as in Proposition 7, provided W_N is retained or $e^{-\kappa n_*}\leq Ct^\alpha$ in the allowed volume/scaling regime. The same estimates for the perturbed actions produced by preceding steps are the additional iteration obligation.

The mass bound in (3) establishes the free reference's locality. The remaining estimate is (18) for the **full normalized interacting integral**, uniformly in the plane size and across the required gauge charts, with W_N defined as the torus kernel minus the periodized bulk kernels and the saddle's uniqueness supplied by $H \geq (4 - C\varepsilon)I$ on the small-field set. Fixed-spin image bounds and a finite-cube Laplace expansion supply local ingredients only; the uniform estimate is the open item. The remainder exponents, $\alpha = \frac{3}{2} - 3\delta$ (strong) and $\frac{1}{2} - \delta$ (weak) with $\delta < \frac{1}{6}$, suffice for one step.

7b. The coefficients for $SU(N)$ and $SU(3)$

Formal one-loop group factor. With the metric $\langle X, Y\rangle = -2\,\mathrm{tr}\,XY$ used throughout, the order-t coefficients (1) for $G = SU(N)$ are those of $SU(2)$ multiplied by $N/2$:

$$\delta_t^{12} = \delta_t^{13} = \frac{Nt}{48}\int_{\mathcal{B}} \frac{q}{R} + O(t^2), \qquad \delta_t^{23} = \frac{N}{2}\,t\Big(-\frac{1}{12} + 2\mathcal{I}\Big) + O(t^2).$$

For $SU(3)$: $\delta_t^{12} = \delta_t^{13} = \frac{t}{16}\int_{\mathcal{B}} q/R > 0$ and $\delta_t^{23} = t(-\frac{1}{8} + 3\mathcal{I}) < 0$. The integrals $\int q/R$ and $\mathcal{I}$ of (2) are group-independent, so the signs of §§2–4 carry over.

Proof. The coefficients are quadratic in the background and one-loop, and every group factor in §§1–4 enters through the adjoint action of the background on the fluctuations. Three facts make the count.

(a) For $su(N)$ the Killing form is $2N$ times the trace form, so $\mathrm{tr}_{\mathrm{adj}}(\mathrm{ad}\,X)^2 = 2N\,\mathrm{tr}\,X^2 = -N|X|^2$; for $SU(2)$ this is the $-2|X|^2$ used in (8). The cut shift (8) is linear in $\mathrm{tr}_{\mathrm{colour}}(\mathrm{ad}\,X)^2$, and (6) and the exact bridge Gaussian (7) hold for any compact group (the zero-image factors $\prod_{\alpha>0}\frac{\alpha(\xi)/2}{\sin(\alpha(\xi)/2)}$ of the heat kernel cancel the Haar Jacobian), so δ^{12}, δ^{13} scale by $N/2$.

(b) The mid-face amplitude uses the Haar Jacobian $j(Y) = \prod_{\alpha>0}[\sin(\alpha(Y)/2)/(\alpha(Y)/2)]^2$, whose logarithm is $-\frac{1}{12}\sum_{\alpha>0}\alpha(Y)^2 + O(Y^4)$, and $\sum_{\alpha>0}\alpha(Y)^2 = -\frac{1}{2}\,\mathrm{tr}_{\mathrm{adj}}(\mathrm{ad}\,Y)^2 = \frac{N}{2}|Y|^2$. For $SU(2)$ the sum is $|Y|^2$, which gave $-t/12$; for $SU(N)$ it gives $-\frac{N}{2}\cdot\frac{t}{12}$.

(c) The transverse coefficient is an Ad-invariant quadratic form on the simple algebra $su(N)$, hence a multiple of the Killing form, and it may be computed on a constant Cartan background Y. There (10) splits by root spaces: Q_Y acts on the root space of α as $h_{\alpha(Y)}$, commutators of different root spaces are orthogonal to Y, and the Cartan directions keep the free Hessian. Each root $SU(2)$ is embedded isometrically, with $T_3^{(\alpha)} = \frac{i}{2}(e_{ii} - e_{jj})$ of unit norm and charge 1 on its root space, so each positive root contributes one complex charged field with the Hessian (11) at $B = \alpha(Y)$. The B^2 coefficient of $\log\det(8 + K)$ therefore adds up to $\sum_{\alpha>0}\alpha(Y)^2 = \frac{N}{2}|Y|^2$ times the $SU(2)$ value per unit B^2. With (b), δ^{23} scales by $N/2$. □

For a general compact simple group the factor is the ratio of adjoint Casimirs in the same metric (the referee notes that each root plane depends on $\alpha(Y)$ alone, so this holds for non-simply-laced groups too), the one-loop pattern that also gives $b_0 = 11N/(48\pi^2)$ in the four-dimensional note. The scaling covers the bulk coefficients only. The dimension-six content of $SU(3)$ adds the d_{abc} cubic noted in §6, and the finite-torus winding term (16) depends on flat holonomies and $\mathbb{Z}_3$ sectors, which the small-field note, §16, treats separately.

8. Consequence for STATE

Atlas cell 2 now has a formal full-plane relative-order-t calculation: the cut couplings shift positively, the transverse coupling negatively, and all other bulk smooth-field terms start at dimension six. The winding term (16) obstructs P(α) as literally stated at fixed plane size N; in

the refinement scaling it lies in the remainder, and P(α) gains the clause $\min_j N_j \geq c_0/t$. The next obligation for this cell is the normalized estimate (18), with winding control, followed by stability under iteration. The infrared spectral-gap obligation remains separate. For $SU(3)$ the bulk coefficients are the $SU(2)$ ones times $\frac{3}{2}$ (§7b). STATE continues to route work through the atlas; the coefficients and their derivation live here.